

NAME: Abhipsa Chakraborty

ENROLLMENT NUMBER: A91704118039

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EXP-4

TITLE: Finding primer from primer 3

AIM: To design primers using primer 3

INTRODUCTION: Primer3 is a computer program that suggests PCR primers for a variety of applications, for example to create STSs (sequence tagged sites) for radiation hybrid mapping, or to amplify sequences for single nucleotide polymorphism discovery. Primer3 can also select single primers for sequencing reactions and can design oligonucleotide hybridization probes. Primer 3 is a widely used program for designing PCR primers.

Gene tool is a custom software that allows the T_m formula $T_m = (wA + xT) * 2 + (yG + zC) * 4$

Where w,x,y,z are the number of the bases A,T,G,C in the sequence respectively.

PROCEDURE:

- We will first visit the NCBI gene homepage then we will type cell division protein FstZ.

- A page will open, we have to click the *Escherichia coli* species and then we will click on FASTA and the FASTA format will get opened.
- Then we will search for primer 3 and will click on primer 3 input version (0.4.0).
- We will copy that FASTA sequence and then paste in that primer 3 webpage in the dialog box provided for pasting sequence.
- We will adjust the parameters like product size ranges from 50-150, Maximum Tm difference will be 5, Primer size will be maximum 24 and minimum 18, Primer GC% will be maximum 60 and minimum 40 and then will click on pick primer and a page will open with primer output.
- We will open the GC% DNA calculator and paste that FASTA sequence and we will get content of all the bases and also the GC%.
- From the page of cell division protein FtsZ we will now go to *Mycobacterium tuberculosis* species then again; we will open the FASTA format of the sequence.
- We will visit the reverse complement homepage and calculate the reverse complement of the FASTA sequence.
- We will open the primer 3 input page and paste the new FASTA sequence of *Mycobacterium tuberculosis* in the dialog box available. Then we will adjust the parameters like product size ranges from 50-100, Maximum Tm difference will be 5, Primer size will be maximum 24 and minimum 18 and optimum 20, Primer GC% will be maximum 80 and minimum 20. We will select the first primer from FASTA sequence of *Mycobacterium tuberculosis* and paste in pick left primer or use left primer text box and the select the first primer of the reverse complement of the FASTA sequence of *Mycobacterium tuberculosis* and paste in pick right primer or use right primer text box.
- A set of PCR primer is shown as an output.

RESULT: The result shows the length, melting temperature, GC content, product size, length. The result is given in the form of sequences for the forward and reverse primers. The pictures pasted shows all the intermediate steps performed all with the final result that show PCR primer designed from primer 3.

Primer3 Output

No mispriming library specified
Using 1-based sequence positions

WARNING: Left primer is unacceptable: Tm too high/Long poly-X; Right primer is unacceptable: Tm too high/High 3' stability

SEQUENCE SIZE: 1140
INCLUDED REGION SIZE: 1140

```

1  ATGACCCCCCGCACAACTACCTGGCGCTCATCAAGGTCGTGGGTATCGGTGGTGGCGGT
61  GTCAACGCCGTCACCGAATGATCGAGCAGGGCCCTCAAGGGGTGAATTCATCGCGATC
121  AACACCGAGCCGAGGCTTGTGATGAGCGATCGCGAGCTCAACTCGAGCTGGCCGC
181  GACTCCACCCGCGGGCTGGGGCCCGCGCGGATCCGAGAGTCGGCGTAAGGCCCGCCGAG
241  GACGCCAAGGACGAGATGAAGAGCTGCTGCGCGGTGCGGACATGGTGTGTTGTCACCGCC
301  GCGGAGGGGGGCGGAGACCGGACCGGGGGGGACCGCTGCGCGCAGCATCGCCCGCAAG
361  CTGGGCGGCTTGACCGTGGTGTGTTGTCACCGCGCTTCTGTTTCGAGGGCAGCGAGCG
421  AGCAATCAGGCGCAAAATGCGATCGCGGCGCTGCGGAGAGTTGCGAGACCTCATCTG
481  ATTCCCAACGACCGGTTGCTGAGATGGAGATGCGCGGATATCGCTGATGATGCTTTC
541  CGTAGCGCCGACGAGGTGCTGCTAACGGCGTGCAGGGGATCACGACCTGATTACACCC
601  CCGGGTCTAATCAACGTGACCTTCGCGAGCTCAAGGGCATCATGTCGGTGGCGCGACCC
661  GCAGTATGGGATCGGCTGGCGCGGGCGAGGGCGGCTGCTCAAGCGGCGGAGATC
721  GCCATCAACTCGCGCTTGTGAGAGCTCGATGGAGGGCGCGCAAGGCGTGGTGTG
781  ATCGCGCGGCGGAGGACTTGGGCTTGTGAGATCAACGAGGCGGCTCGTTGGTACAA
841  GACGCGGCTACCGCGATGCAACATCATCTTCGCGACGCTGATCGAGGATTCGCTCGGT
901  GACGAGGTGCGGTTGACGTGATCGCGCGGTTGACGCTAGGCTCCGCGCGCGAG
961  CGGCTGATGGGCGAGACCGCGCGCGCCACCGGATGAGTACGCGAGGCGAGGCAAGCTC
1021  ACCTGACCTTGTTCGAGCGGTCGACGCGCTGAGCGTGGCTGTCACACCAACGGCGCA
1081  ACCCTGAGCATCGGGGATGACGAGGATGTCGAGCTGCCCGCTTCATGCGCGCGTGA

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No Acceptable Primers Were Found

The statistics below should indicate why no acceptable primers were found. Try relaxing various parameters, including the self-complementarity parameters and max and min oligo melting temperatures. For example, for very A-T-rich regions you might have to increase maximum primer size or decrease minimum melting temperature.

Statistics
Pair Stats:

This figure shows the output produced by providing right primer and left primer sequence

Reverse Complement

Reverse Complement converts a DNA sequence into its reverse, complement, or reverse-complement counterpart. You may want to work with the reverse-complement of a sequence if it contains an ORF on the reverse strand.

Paste the raw or FASTA sequence into the text area below:

```

CGGGCCCGACCGGATCGAGTCAGCGAGGCGAGGTCACCTCGACCTTGTTCGAG
CGGTCGAGCC
GTACGCTGCGCTTGCACACCAAGCGCGCAACCTGAGCATCGCGGTGATGACGACG
ATGTCGAGCTGC
CGGCTTTCATGCGCGCGCTGA

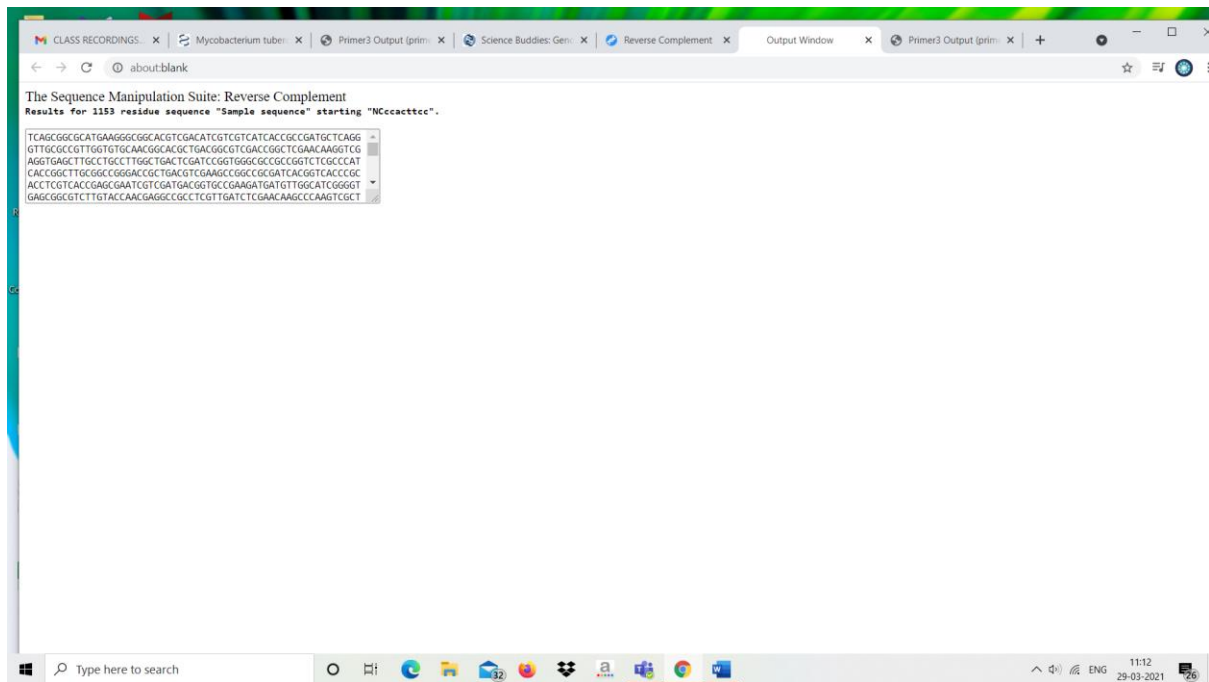
```

SUBMIT CLEAR

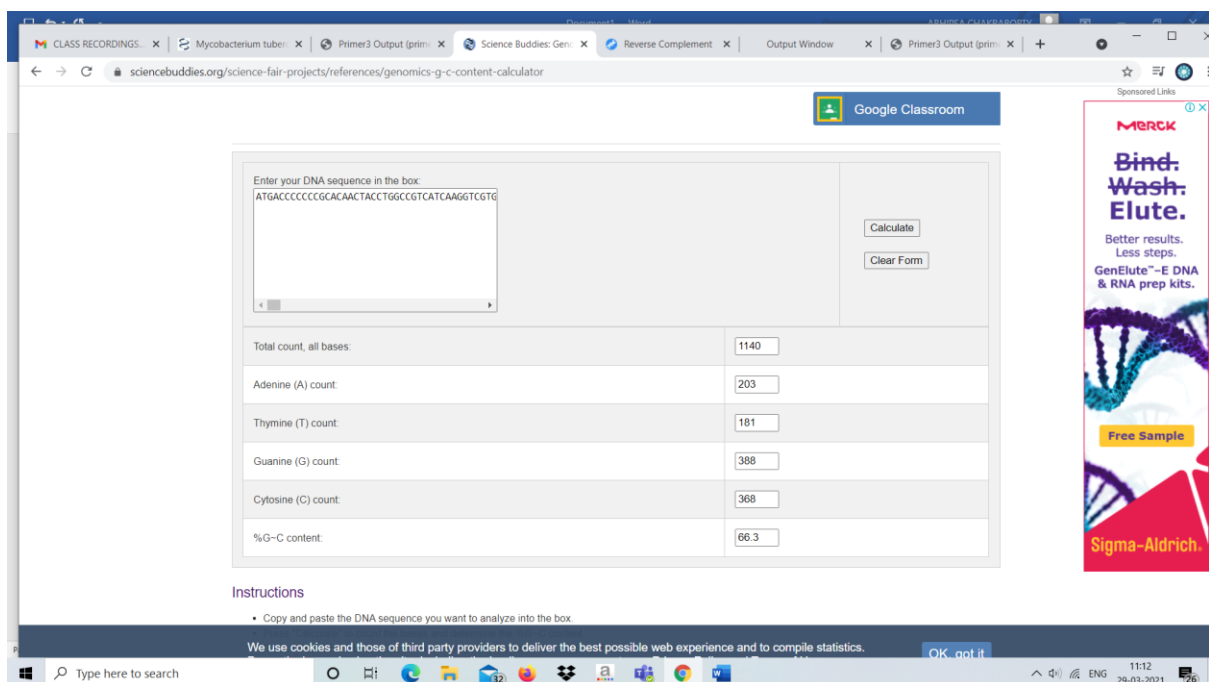
Convert the DNA sequence into its counterpart.

[home](#)

This figure shows the way to calculate reverse complement of the FASTA sequence



This figure gives the result of the reverse complement of the FASTA sequence



This figure shows the calculation of all the bases and the GC% content as well

The screenshot displays the NCBI Nucleotide database interface. At the top, there's a search bar with 'Nucleotide' selected. Below it, a warning banner states 'COVID-19 is an emerging, rapidly evolving situation.' with links to CDC, NIH, SARS-CoV-2 data, and HHS. The main content area shows the 'Mycobacterium tuberculosis H37Rv, complete genome' with the NCBI Reference Sequence: NC_000962.3. The FASTA sequence is displayed, starting with >NC_000962.3:2409524-2409524 Mycobacterium tuberculosis H37Rv, complete genome. On the right, there are options to 'Change region shown' (Whole sequence or Selected region) and 'Customize view' (Display options: Show reverse complement, Show gap features). Below these are links to 'Analyze this sequence' (Run BLAST, Pick Primers, Highlight Sequence Features, Find in this Sequence) and 'Related information' (Assembly).

This figure shows the FASTA sequence of *mycobacterium tuberculosis*

The screenshot shows the 'Primer3 Output' page from bioinfo.utel.pt. It displays the results of a primer design run. Key information includes: 'No mispriming library specified', 'Using 1-based sequence positions', and 'OLIGO' details (start, len, tm, gc%, amp, 3' seq). The 'LEFT PRIMER' is 370 20 60.00 55.00 3.00 1.00 GGATATCCTGACCGTGTGCTG and the 'RIGHT PRIMER' is 510 20 60.03 45.00 2.00 TTTGACGATTTGTGCTG. The 'SEQUENCE SIZE' is 1152 and the 'INCLUDED REGION SIZE' is 1152. The 'PRODUCT SIZE' is 141, 'PAIR AMP COMPL' is 5.00, and 'PAIR 3' COMPL' is 0.00. The sequence is shown with the primers highlighted in red and green. The sequence starts with 1 ATGTTTGAACCAATGGAACCTACCAATGACGCGGTGATTAAAGTCATCGGCTCGGCGGC and ends with 601 ACTGTCGGGTTTGAATGAAGCTGGACCTTTGACAGCTACGACCGTAATGCTGAGATG.

This figure shows the set of PCR primers developed by primer 3

The screenshot displays the NCBI Nucleotide database interface. At the top, there's a search bar with 'Nucleotide' selected. Below it, a COVID-19 warning banner is visible. The main content area shows the FASTA sequence for 'Escherichia coli str. K-12 substr. MG1655, complete genome'. The sequence is displayed in a monospaced font. On the right side, there are several interactive elements: a 'Change region shown' section with radio buttons for 'Whole sequence' and 'Selected region', a 'Customize view' dropdown, and a 'Related information' section with links to 'Assembly', 'BioProject', 'BioSample', and 'Protein'. The bottom of the page shows a Windows taskbar with various application icons and a system clock indicating 11:15 on 29-03-2021.

This figure shows the FASTA sequence of *Escherichia coli*.

CONCLUSION: From these pictures and observations we can conclude that primer 3 is a powerful bioinformatic tool for designing sets of PCR primer.